

International Workshop on Energy Storage in the Grid

Low, Medium and Large Scale Requirements

Barcelona, 8th - 10th January 2014

EV-Grid Concept

Electrical Architecture & Technologies, research tools

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IFSTTAR

French institute of science
and technology for transport,
development and network



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Urban transportation mission profile,...Energy production profile...
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EV today's, Recharging station and ,...
4. Reliability & Safety, End of life
Calendar, Power cycling, Thermal cycling, ...short-circuit fault
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6. Synthesis

Synergy between Renewable Energy and Transportation Systems

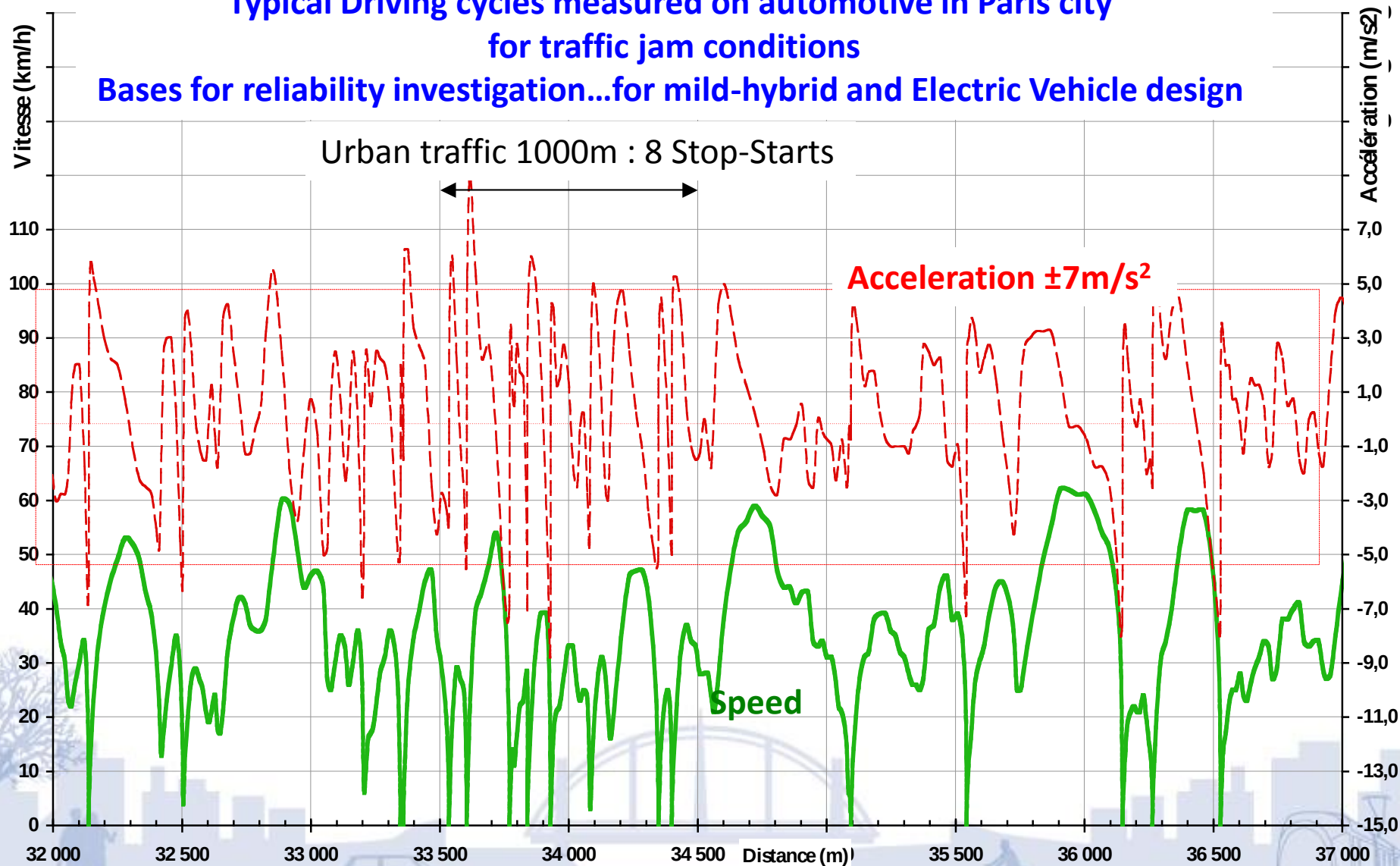
- Reduce energy consumption and pollutant emission
- Increase efficiency, reliability, durability and service quality

Why Energy Storage System

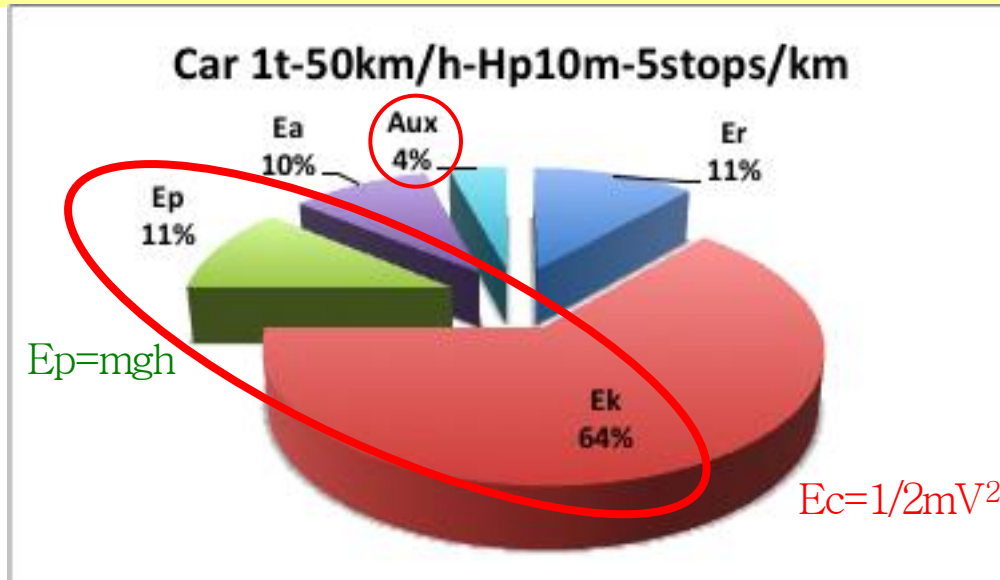
Automotive driving in urban mission profile

Typical Driving cycles measured on automotive in Paris city
for traffic jam conditions

Bases for reliability investigation...for mild-hybrid and Electric Vehicle design



Energy in City Mobile applications: Kinetic & Potential Energy



Car energy consumption
in traffic city:

120Wh/km if 1 stop/start/km

244Wh/km if 5 stop/start/km

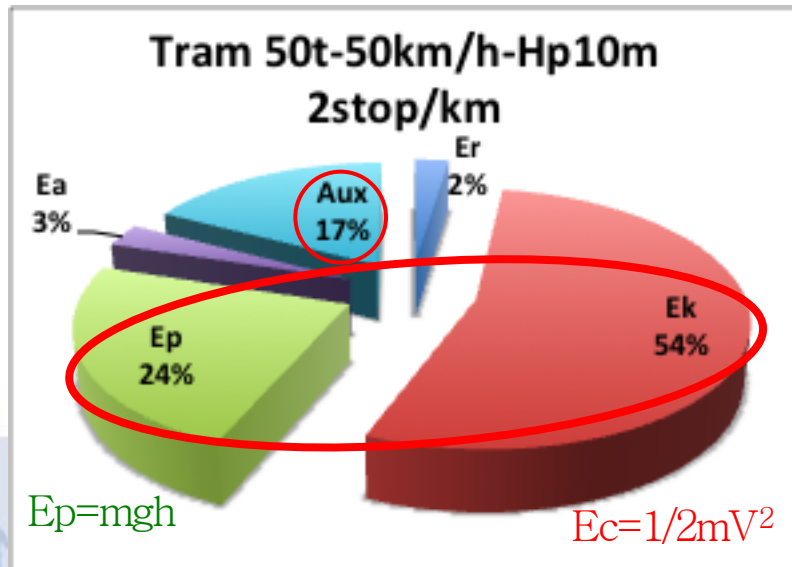
50 to 75% can be saved in theory

30 to 50% could be in practical

by using regenerative braking

(Leaf EV data: 155-212Wh/km)

Energy consumption: $\sum E = R.V + D.V^2 + \frac{1}{2}mV^2 + Mgh + \text{Aux. \& Comfort}$



Tramway energy consumption
in traffic city:

5,8kWh/km if 2 stop/start/km, 10m elevation

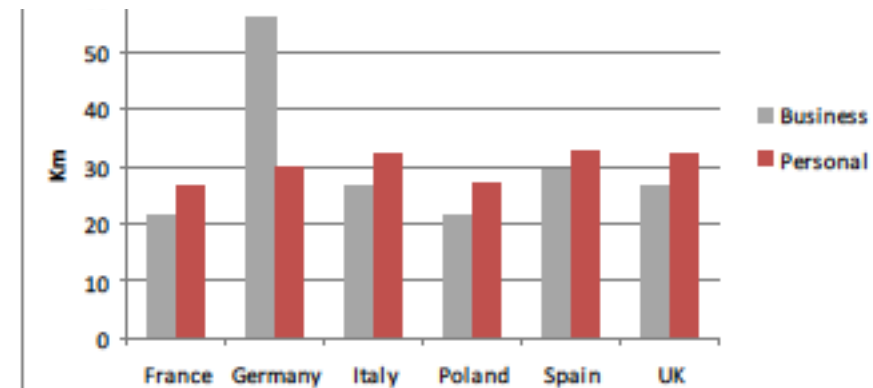
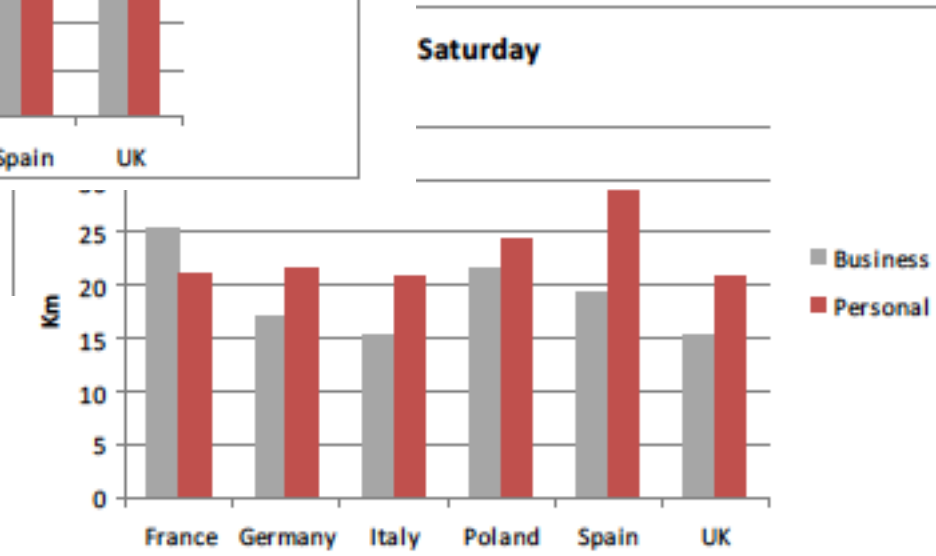
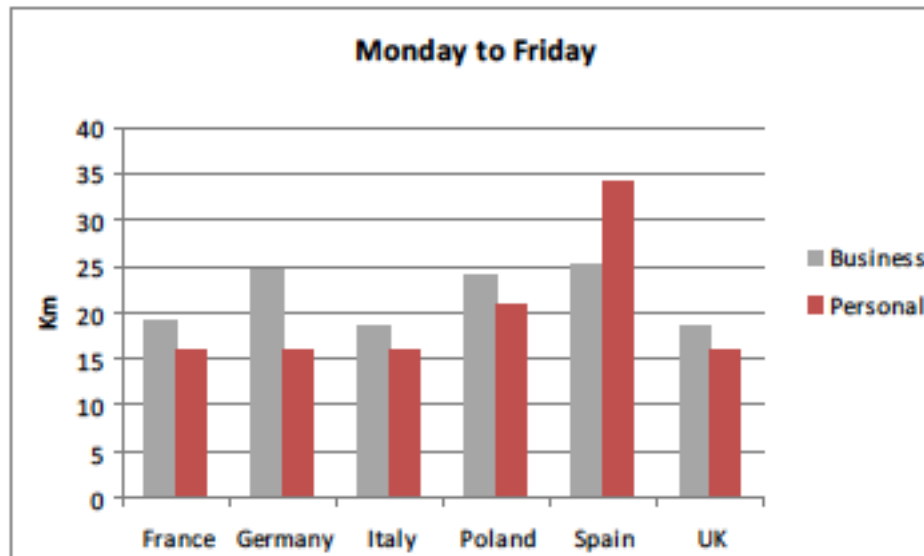
75% can be saved in theory

about 40% in practical

by using regenerative braking

Average trip distance (kms) By trip purpose

Source: Driving & parking patterns of EU drivers a mobility review, JRC-EU2012



Distribution Trips / day

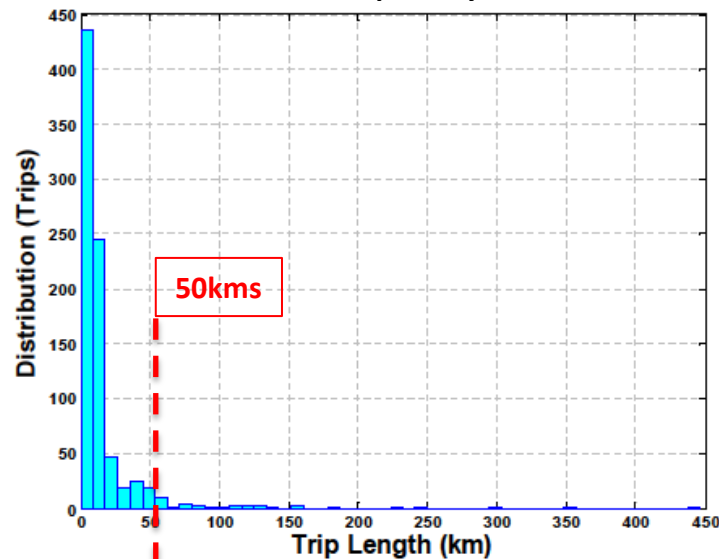
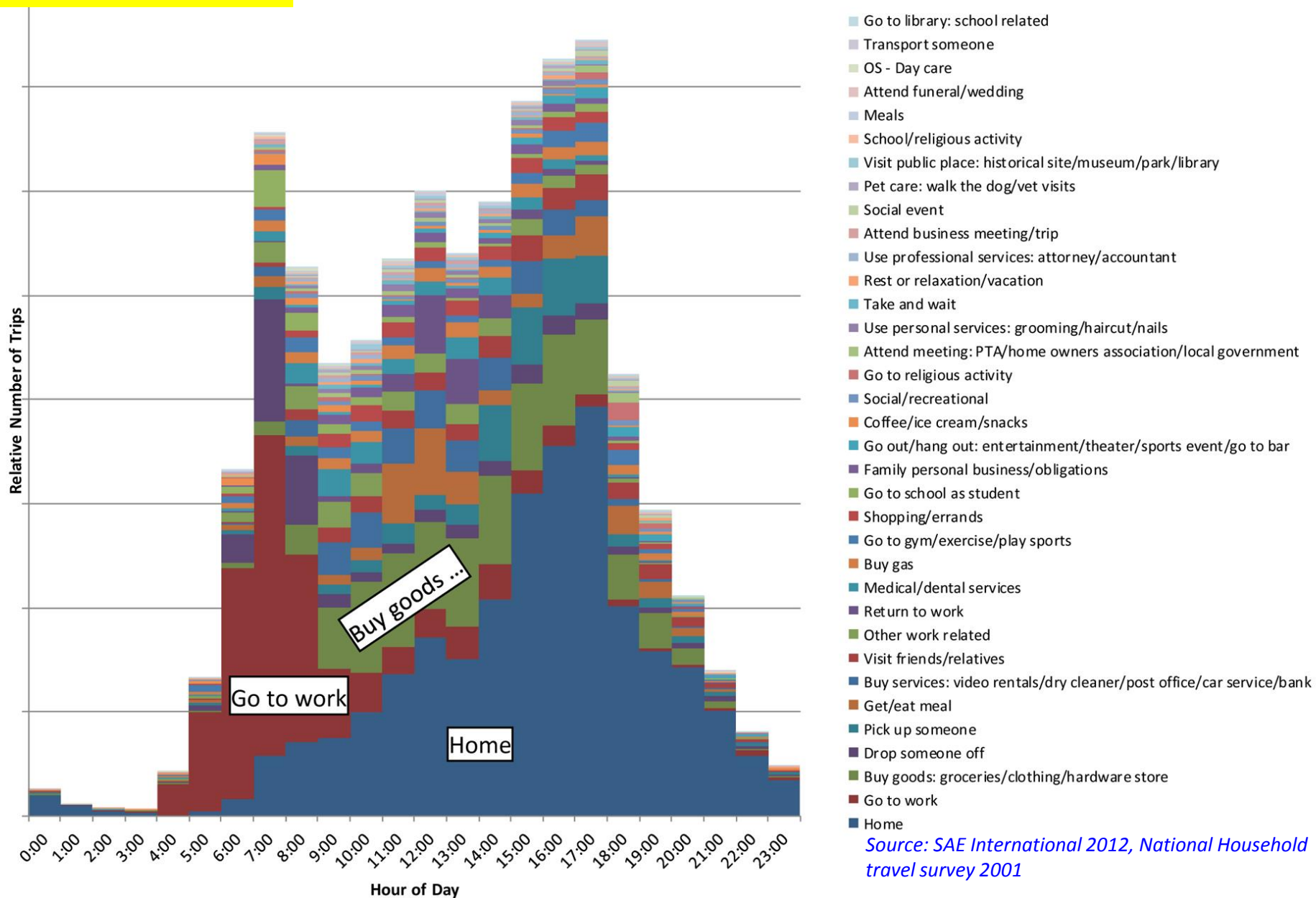


Figure 88: Distribution of trip lengths among the CarChip-logged data by trip
Source: I. Berry, Thesis-MIT-USA2010,
National Household travel survey 2001

How People Use Their Vehicles

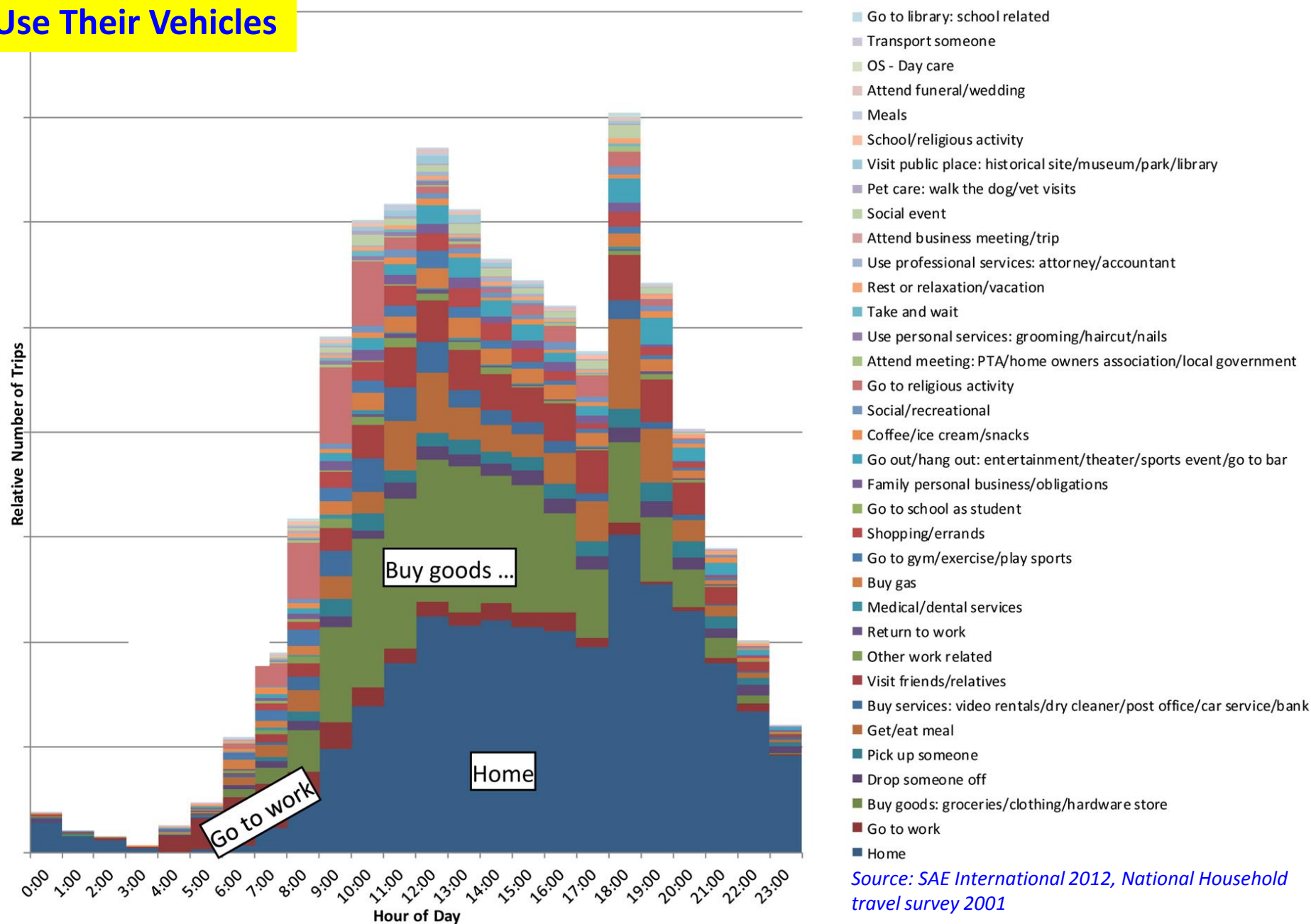
Destinations vs. Time of Day (Weekdays)



Source: SAE International 2012, National Household travel survey 2001

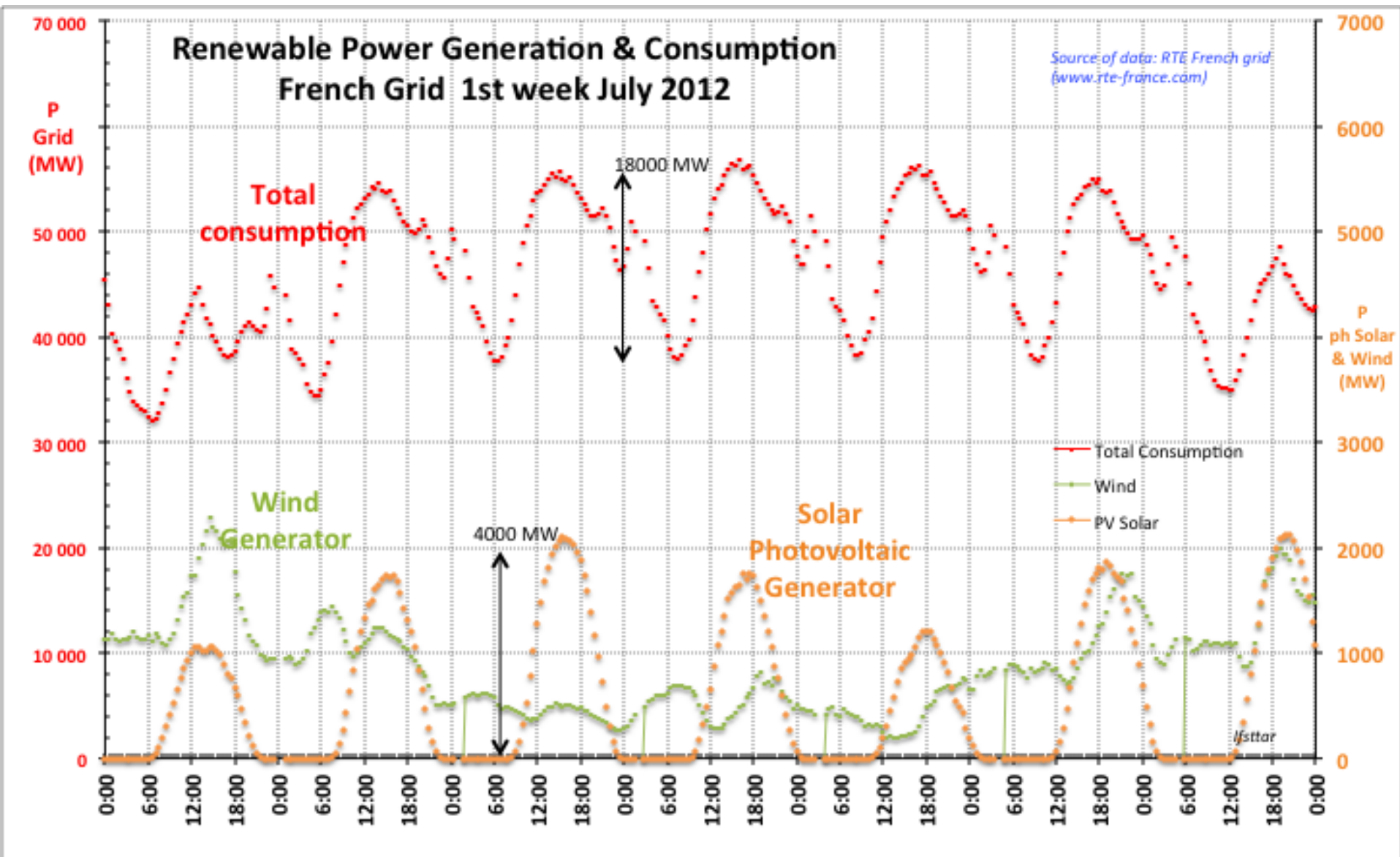
How People Use Their Vehicles

Destinations vs. Time of Day (Weekends)



Source: SAE International 2012, National Household travel survey 2001

Renewable energy, uncertain production and time gap between consumption and production require Energy Storage System



Why Energy Management from generation to electrical vehicle: traction and renewable conditions

- sizing of generator and infrastructure on average power: “down sizing”
- manage energy from renewable source to EV and to GRID
- reduction of consumption: regenerative energy

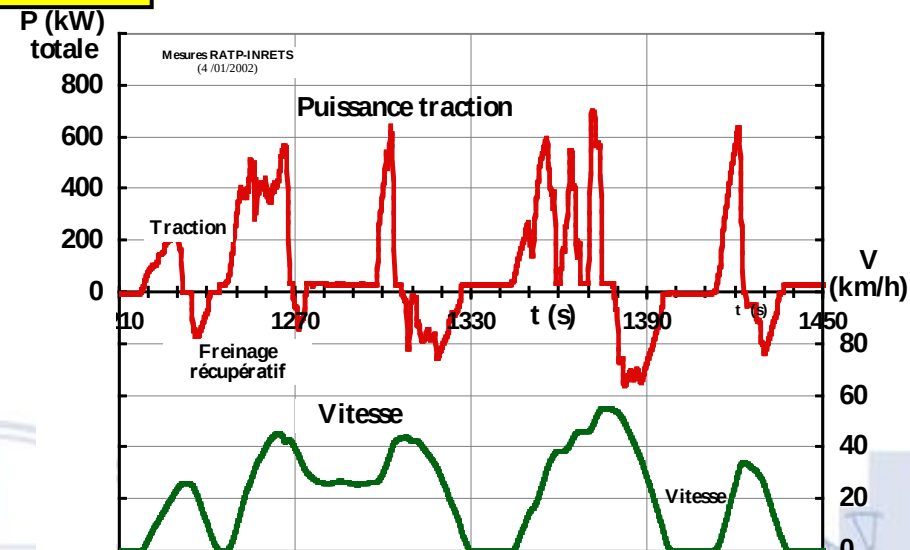
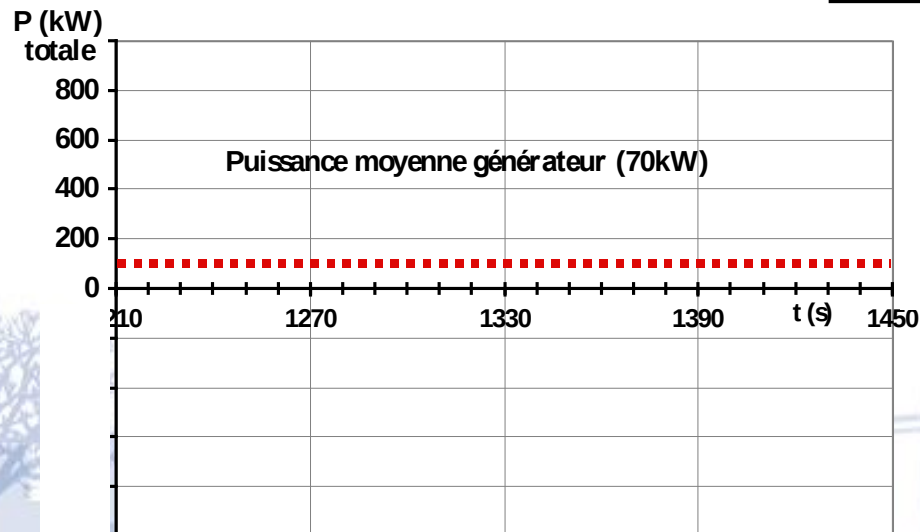
Average Power Used for moving
(ex. tramway : 70kW)



**Buffer
Energy
Storage**



Traction peak power
(ex. tramway: 800kW à 1MW)



“Vehicle electrification Level...today status“

From ICE to EV, impact on energy storage technology

Rated Electrical Parameters	Stop Start	Mild Hybrid	Full Hybrid	Recharg. Hybrid	Full Electric	Electric Range Extender
Energy storage	No 12V battery Reinforced Or SCap	100-150kJ 0,04kWh SCap?	1-2kWh	4-5kWh	20-30kWh	16kWh
Electrical Power	2kW peak	10-15kW	60kW	60kW	80kW	111kW
Vehicle Example	BMW, VW PSA, Nissan ...	48V German Carmakers Agreement 2011	1,8kW (NiMH) Prius 2002 0,94kWh Insight	4,4kWh Prius3 2011 27kW-80kg 3kWh-25kms-85kmh	24kWh (li-ion) Leaf 90kW-	16kWh (li-ion) Ampera -198kg -60kms
Prospective Market 2020 <i>Source: actus ?</i>	? ≠30%	? ≠10%	? ≠10%	? ≠10%	? ≠1-2%	?

Source: NREL, Toyota,...

To remember the market : 19Mcars-EU27 2011, 80Mv-World

Electric vehicles, today status !

On board energy storage and recharging power

Vehicle	Class	Battery (kWh)	type	Fast recharging	Motor (kW)	type	Weight No-load
Smart fortwo	2 seats	17,6	Li-ion-wc*	AC-22kVA	55	PM	1000
Fiat 500e	Mini-compact	24	Li-ion-wc	AC-7kVA	83		1000
Mitsub. i-MiEV	Compact	16	Li-ion-wc	DC-50kW	47	PM	1000
Nissan Leaf	Compact	24	Li-ion-la-wc	DC-50kw	80 / 109	SM	1474
Tesla S (85kWh)	Large car	85		DC-120kW			2108
Tesla S (60kWh)	Large Car	60		DC-120kW			
Honda Fit/Jazz	wagon	20	Li-ion-wc		100	PM	
Ford Focus e	Sub-compact		Li-ion-wc		107		1473
Renault ZOE	Compact	25	Li-ion-wc	AC-43kVA	65 / 88	Srwwired	1562
Bolloré BlueCar	Compact	30	LMP	AC-7kVA	45		1120
Smith-Newton	Truck-Bus	80		AC-43kVA	120	Induction	4432/11990
	Truck-Bus	120		AC-43kVA	120	Induction	4432/11990

*wc: water cooled assisted

Urban Energy consumption close to 100Wh/km

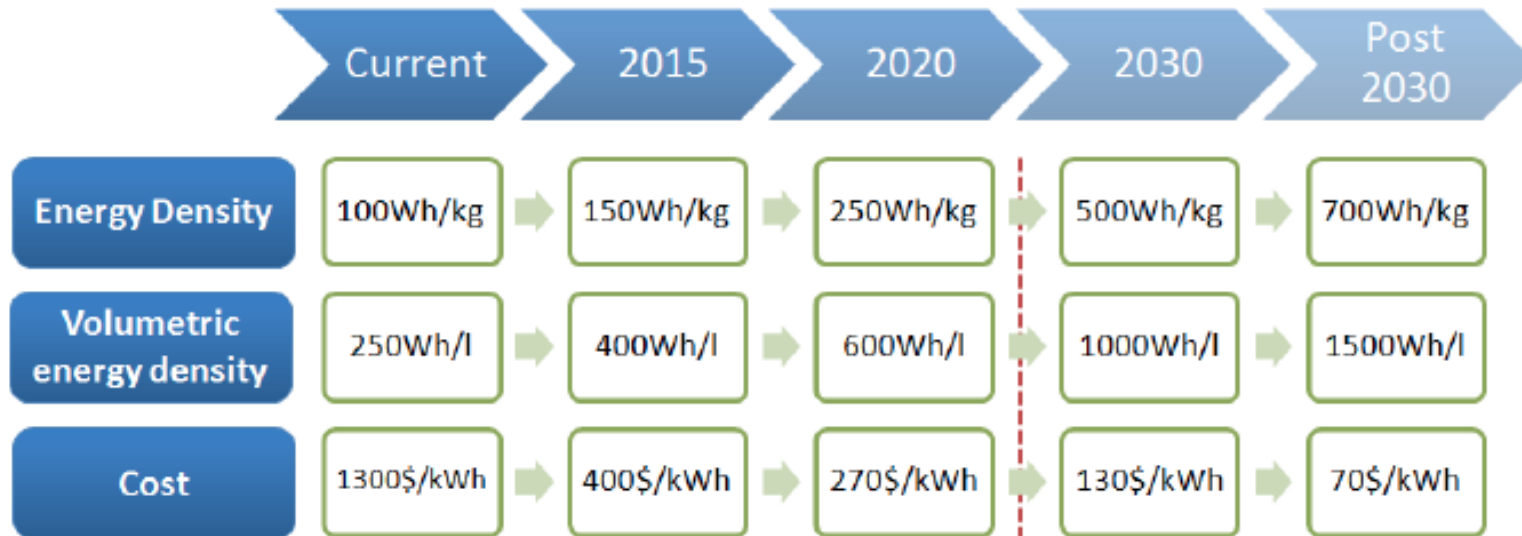
Electric Vehicle Batteries: state of the art cathode chemistry

(Total 2011, about 29GWh produced including phone, labtop,...)

Source: Avicenne 2011

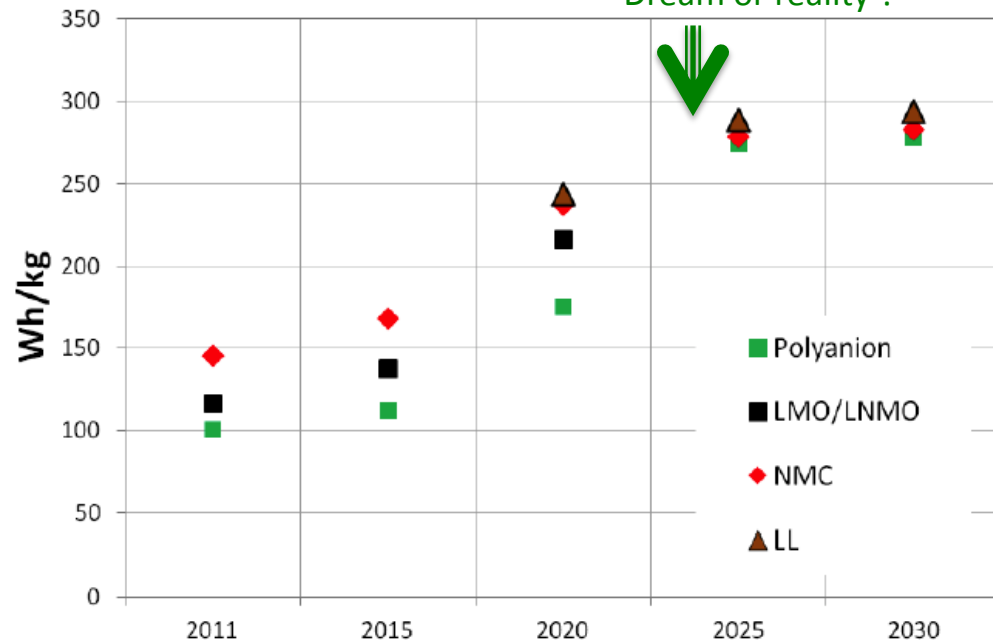
Cathode active material name	Material	Abbreviation	Short form	Maturity for EV?
Lithium cobalt oxide	LiCoO_2 (60%Co)	LCO	Li-cobalt	Used in the original Tesla car but rejected on safety concern grounds by series car OEMs
Lithium manganese oxide	LiMn_2O_4	LMO	Li-manganese or spinel	Already in series car (e.g. Leaf, Volt, iMiEV)
Lithium iron phosphate	LiFePO_4	LFP	Li-phosphate	Already in series car (e.g. Fisker EV)
Lithium nickel manganese cobalt oxide	LiNiMnCoO_2 (10-20% Co)	NMC	NMC	Used in consumer goods and EV prototypes
Lithium nickel cobalt aluminium oxide	LiNiCoAlO_2 (10-20% Co)	NCA	NCA	Already in series car (e.g. plug-in Prius)

Electric Vehicle battery expected road map



New Technologies

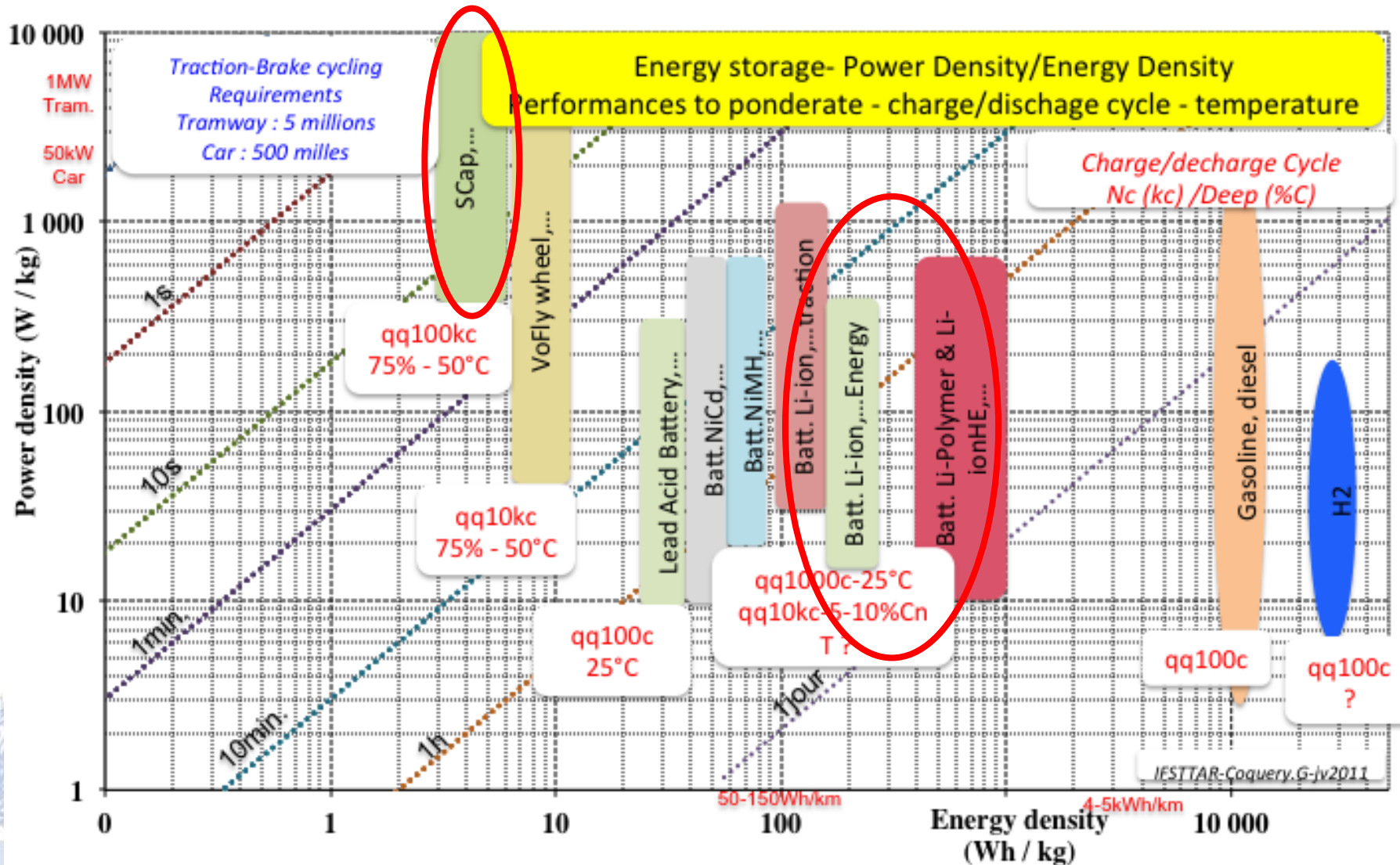
50kWh
400kms autonomy
Dream or reality ?



Other chemical solutions are investigated
NMC, LMO,...

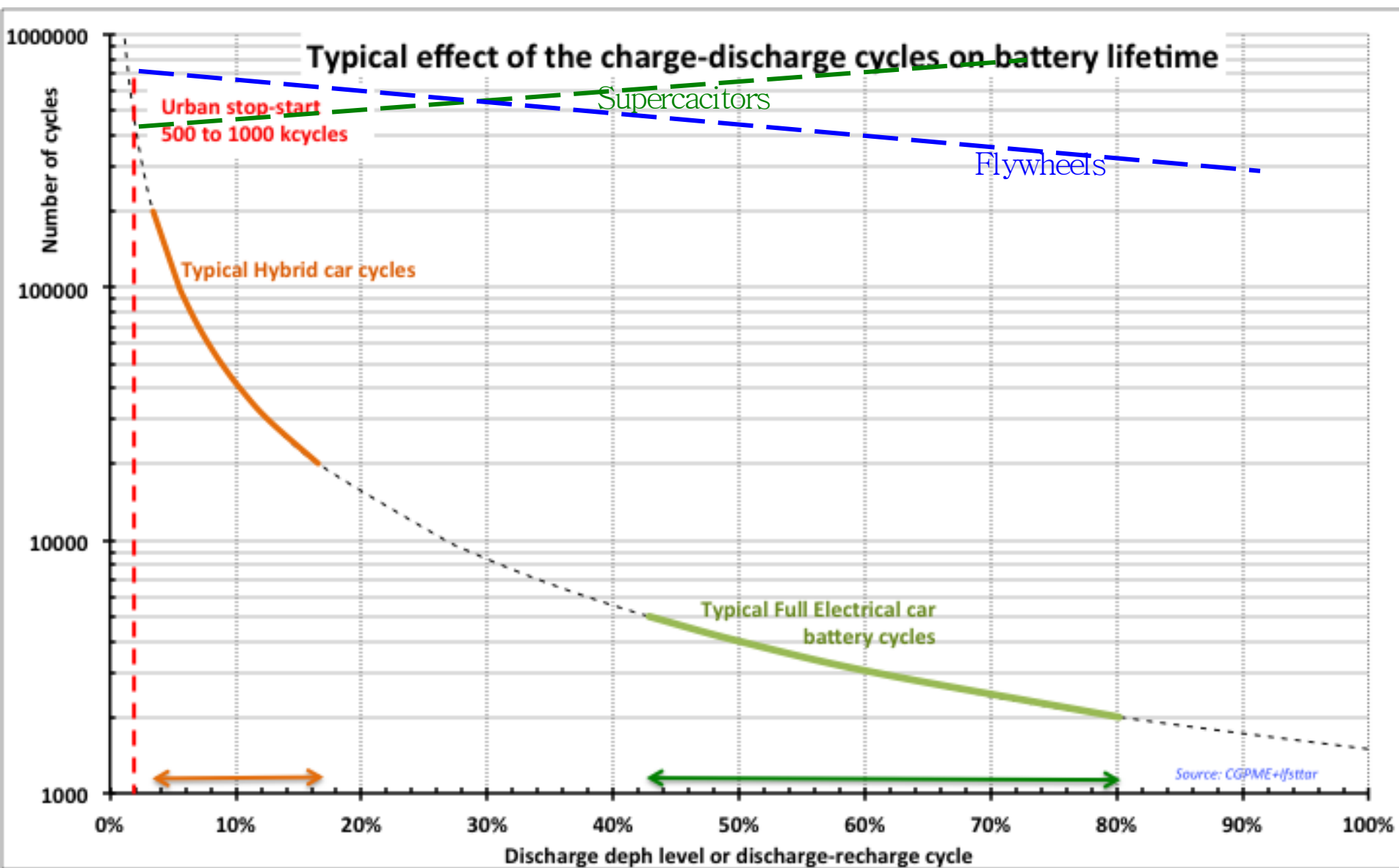
Source: French Embassy Japan 2010

Battery/Supercapacitor – Energy/Power – Efficiency/Reliability Status of the art



Choice of the energy storage technology

Impact of charge-discharge cycles...on battery



Methodology for lifetime & reliability characterisation...no reliability model today ?

Summary of the general test plan proposed :

-1 – Initial measurements

- dimensions, weight,

-2 – Static tests or stabilised conditions (vs U, T, I)

- Calendar tests, leakage current under static voltage (or floating conditions)
- DC method : discharge tests
- Z or impedance spectroscopy method

-3 – Dynamic tests or cycling or pulse conditions (vs U, T, P, t)

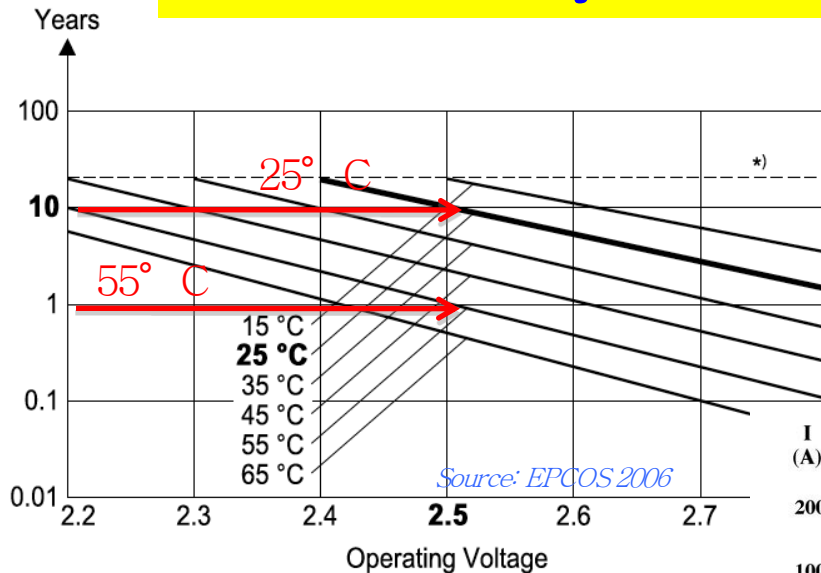
- Thermal cycling test (based on environmental conditions)
- Current cycling tests (based on vehicle mission profile)
- Power cycling tests (repetition of the exact working conditions)
- Short-circuit tests (cell, module, bank with circuit protection)

Monitoring of the test conditions:

- external access : I_{sc} , U_t , T_t , T_c (.t terminal, .c case)
- internal access : U_{ei} , T_{ei} (e_i : electrode i ...)

needs to be install by the manufacturer

Reliability: influence of the working conditions



Lifetime 10 years (at $V_R = 2.5$ VDC and $T_{op} = 25$ °C)

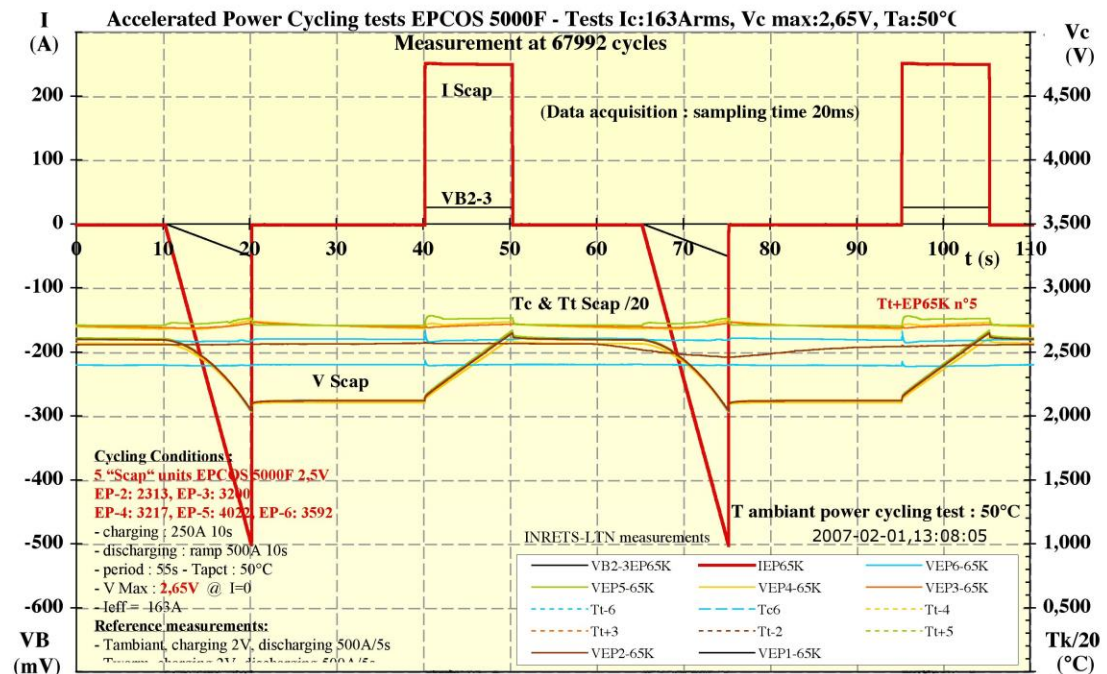
*) Note: The lifetime is shown up to 20 years. An expanded life is additionally influenced by deviations of lifetime parameter to fatigue mechanisms of capacitor materials.

Voltage & Temperature
At waiting status
(calendar ageing)

Homogenous temperature

Eyring model: $t_d(T, U) = A \cdot e^{(U/kT)}$

Voltage, Temperature, Current
During operating time
(cycling ageing)



Non-homogenous temperature
Thermal cycling
Current cycling



WATT system, PVI France

Electromobility: local energy delivering by high power way

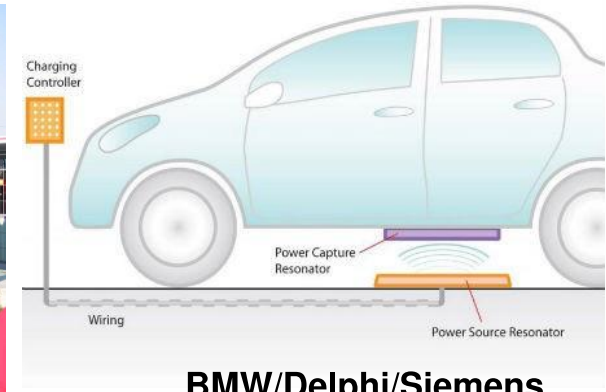
- Fast recharge in stop station
- Dynamic energy transfer



Proterra, USA



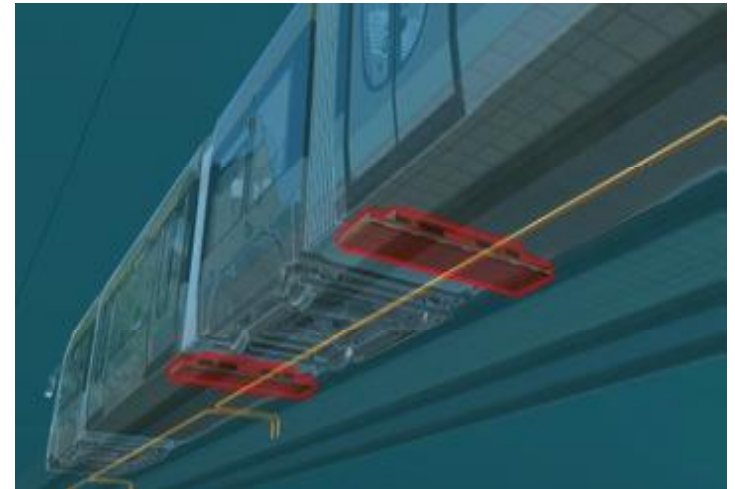
Alstom, Bordeaux, France



BMW/Delphi/Siemens



Capabus, USA/China

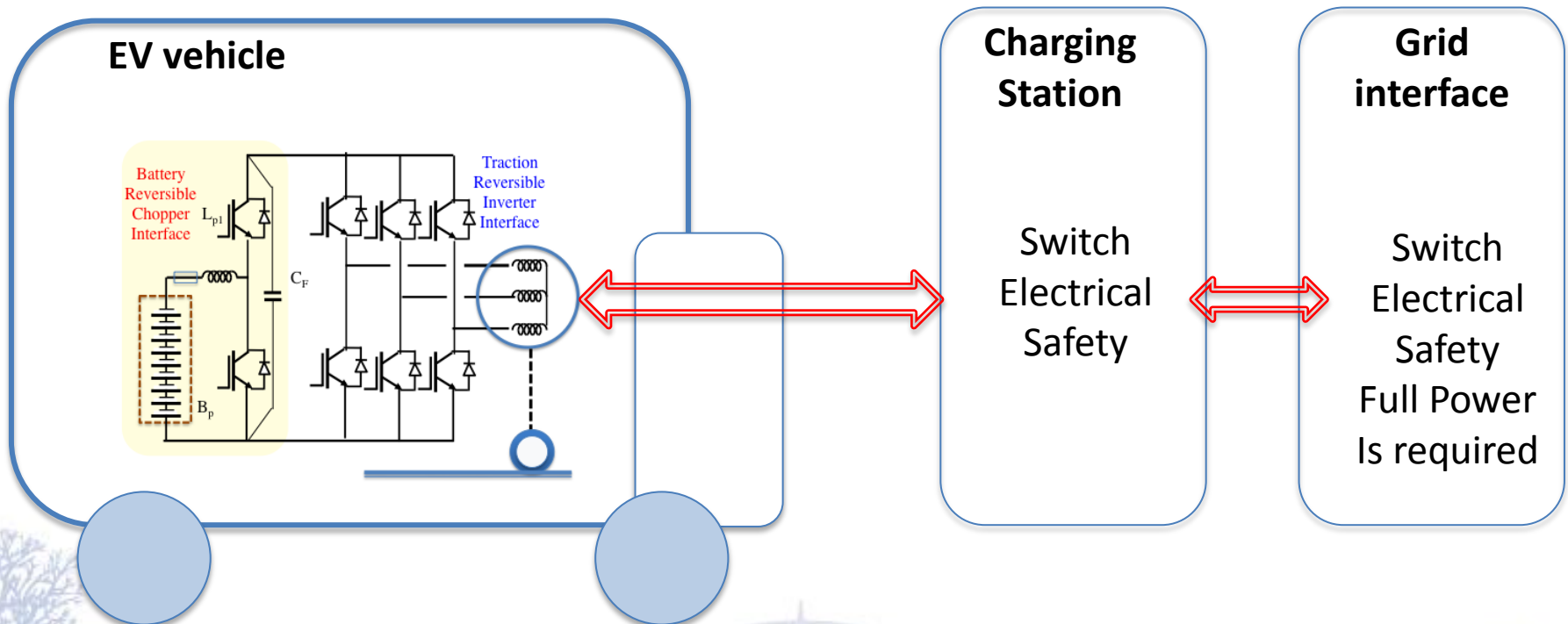


PRIMOVE - Bombardier

Fast charging station to EV

AC 22kVA & 43kVA

The today normal mode is non reversible architecture
Only Grid to EV

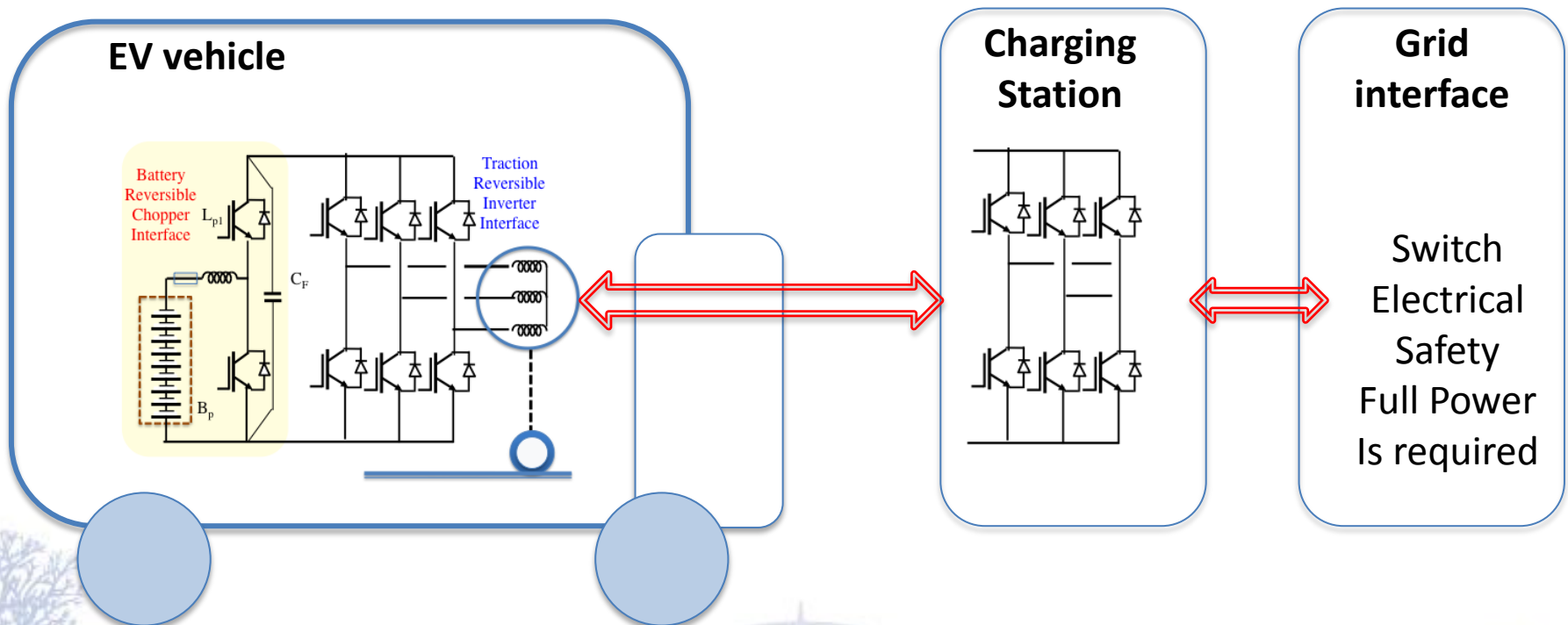


The EV converter assume the respect of the rules on Quality and Safety

Fast charging station to EV

DC 50kW ... 120kW

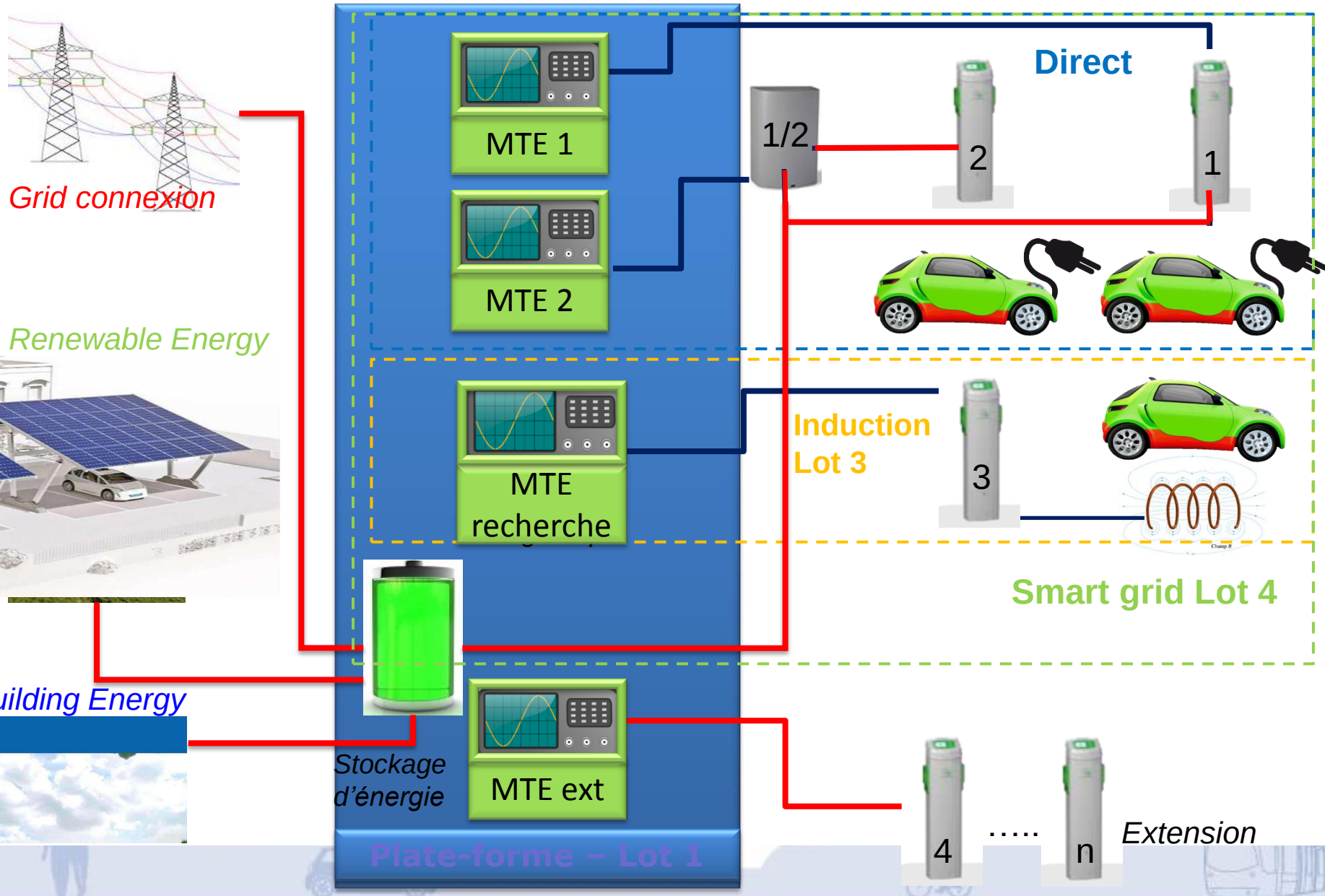
Today the reversible mode is working... Grid to EV to Grid



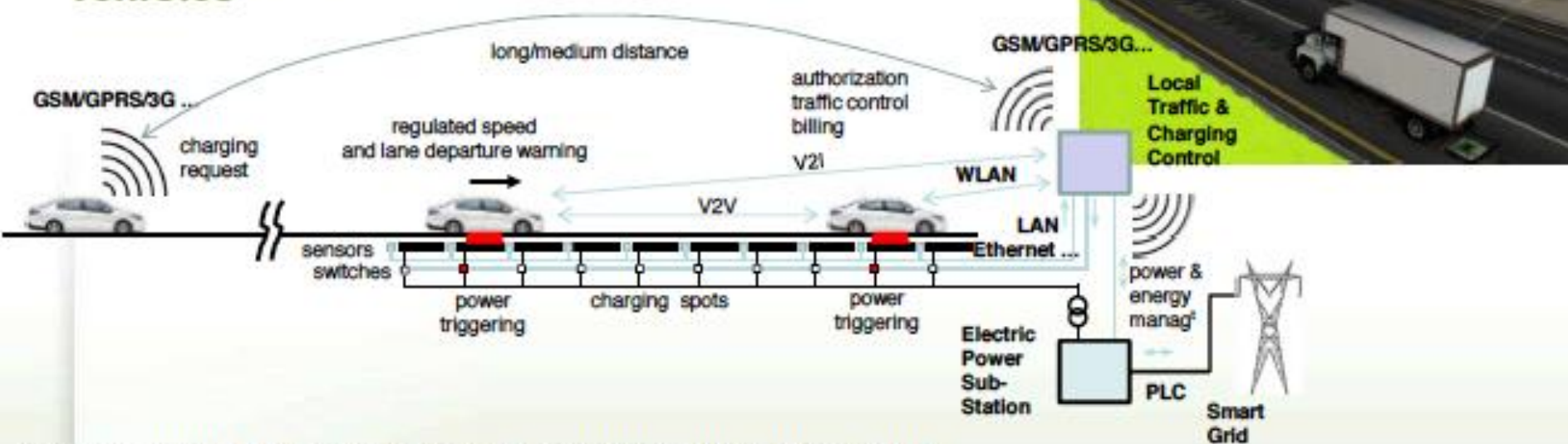
The Charging station converter assume the respect of the rules on Quality and Safety

Platform for test on fast charging station AC & DC

Energy management ...grid - charging station - vehicle



FeAsiBility analysis and development of on-Road charging solutions for future electric vehiCles



- Coordinateur : ICCS (Grèce) ; 26 partenaires dont **VeDeCoM**
- Pays : Italie, Belgique France, Allemagne, Grèce, Pays-Bas, Espagne, Suède, Royaume Uni
- VeDeCoM : Subvention **887 k€** / Coût total : 1182 k€

incluant



PSA PEUGEOT CITROËN

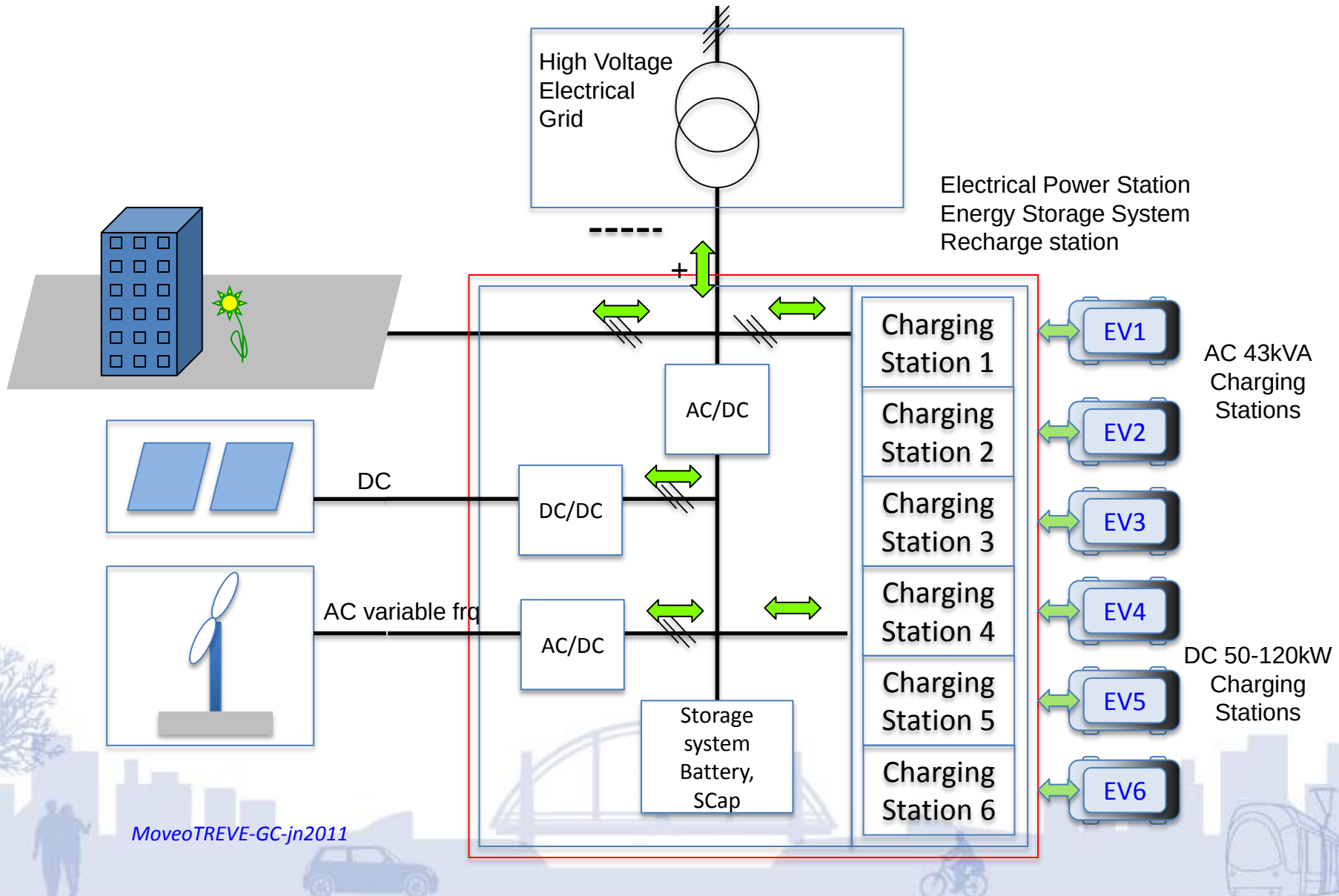


avec



Electrical architecture for EV & HEV Urban Recharge Service Station

Coupling to renewable energy generators



MoveoTREVE-GC-jn2011

EV scenario to help Grid to solve winter peak days demand

Power Δ (Pmax-Pmin) is about 20GW during 24h00

Basic hypothesis...just to discuss

Data based on the French status:

- About 105GW installed, 80 to 95GW required peak power each winter day
- Hypothesis 1 million EV car 2020 ?

Solution1: 25kWh/car... Σ 25GWh...

if free offer for Grid is 10% = 2,5GW during 1h

Solution2: 50kWh/car... Σ 50GWh...

If free offer for Grid is 20% = 10GW during 1h

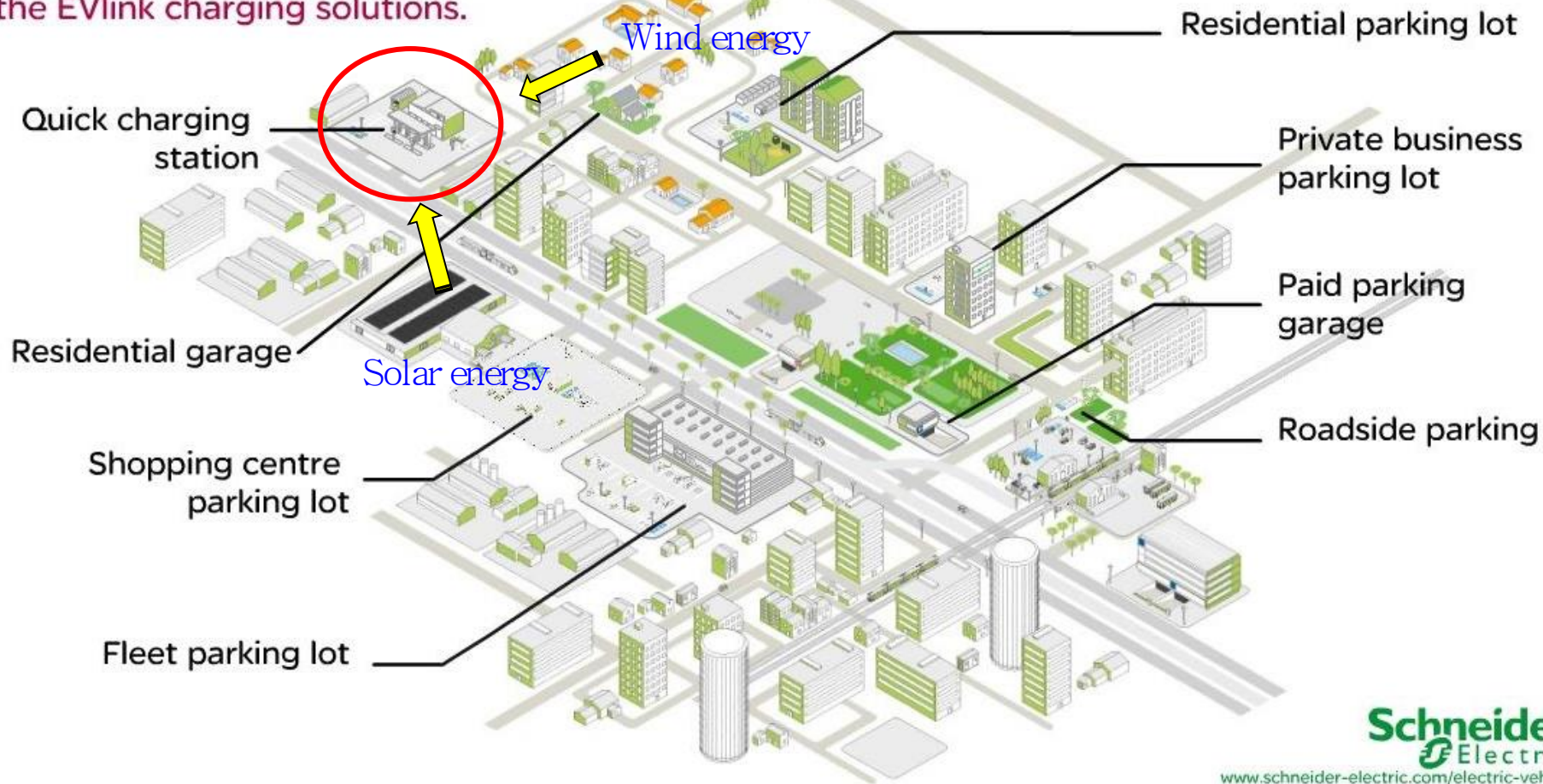
EV storage combined with hydraulic storage could contribute to save the Grid stability and energy delivering



Electromobility in the City: EV Stations-service as energy storage

How to charge an electric vehicle?

Skim through the picture and click to discover the EVlink charging solutions.



solutions recharge batterie voiture vehicules electriques schneider electric

Energy Storage Technologies for EV-Grid concept

Synergy electro-mobility to renewable energy and Grid

Synthesis and prospective

- **Energy status analysis** $\square$ **ratio : Peak power / Average power**
 - > 10 for tramway (without climatic equipt.), > 3 for electrical substation
 - > 3 to 5 due to car stop in urban traffic
 - > Energy renewable production / power needs, PV production 100% day light / 0% night

Electrical buffer storage devices will allow Energy Management

- efficiency improvement (braking regenerative energy, reduction of line losses),
- full electrical braking for LRT and LRV-EV/HEV to save energy and particles emission
- **down sizing of electrical substation and grid connexion**
- partial autonomy for infrastructure cost reduction (free crossroad, existing tunnel,...)

Key technologies and system approach:

- buffer energy storage devices,... **reliability law is the key knowledge to get...**
- integrated semiconductor functions (converter),
- appropriate software and high level data communication between grid and users
- **local energy management in the frame of micro-smart grid**
- **reduce peak power on the grid for charging station**

Efficiency+Reliability+Safety+Cost = Energy storage system required

Thank you for your attention



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Electromobility: Battery interface voltage for traction

Power electronic converter assume :

Voltage-current adaptation for traction

And power sharing between Scap and Battery for highest reliability-efficiency

